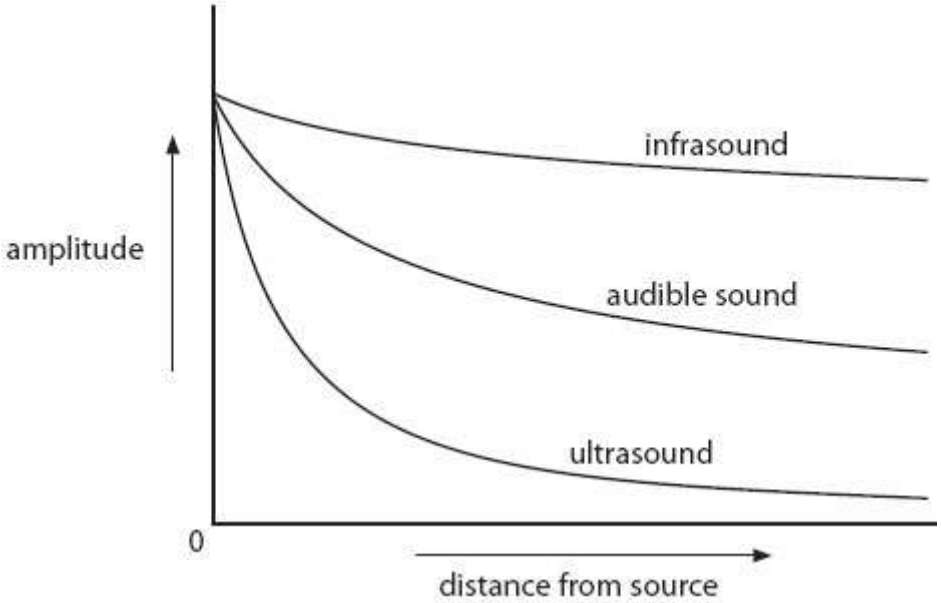
Questions

Q1.

As a sound wave travels through water its amplitude gets smaller.
This graph shows how the amplitude of different types of sound waves decreases as they travel through water.



Whales use sound to communicate over long distances.
Explain which is the best type of sound wave for whales to use when communicating over long distances.

(2)

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.....

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Q2.

Four students and their teacher do an experiment to measure the speed of sound in air.

The teacher stands at a distance and fires a starting pistol into the air.
The students see the flash when the pistol is fired.
They measure the time from when they see the flash to when they hear the bang.

A student drew a diagram of the arrangement as shown in Figure 7.

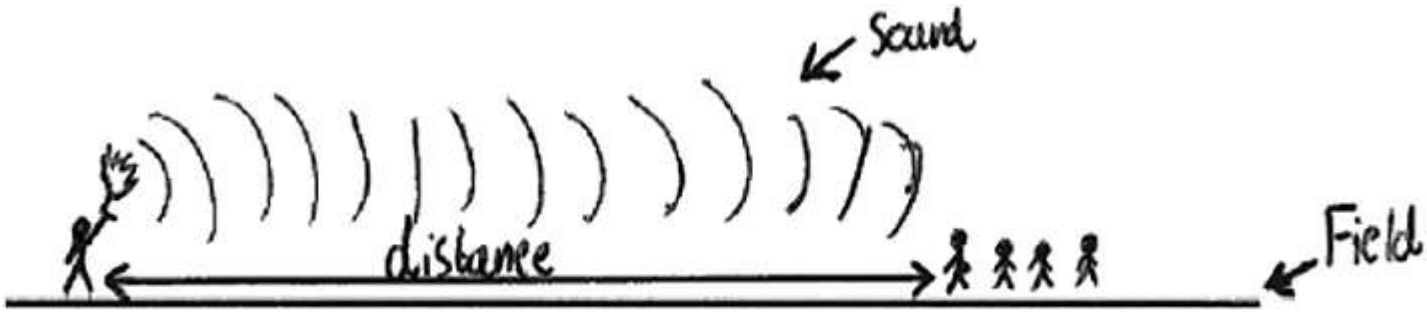

Figure 7

The students obtained a value of 240 m/s for the speed of sound.

The accepted value, in a science data book, is 343 m/s.

(i) Calculate the difference between the students' value and the accepted value as a percentage of the accepted value.

(2)

percentage difference = ... %

(ii) When the distance was 100 m, the students measured the following times:

0.43 s 0.35 s 0.50 s 0.38 s

Explain why their times vary so much.

(2)

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(iii) Explain **one** way the students might improve this experiment.

(2)

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(Total for question = 6 marks)

Q3.

Figure 14 shows the hearing responses of a human, a mouse and a bird over a range of frequencies of sound.

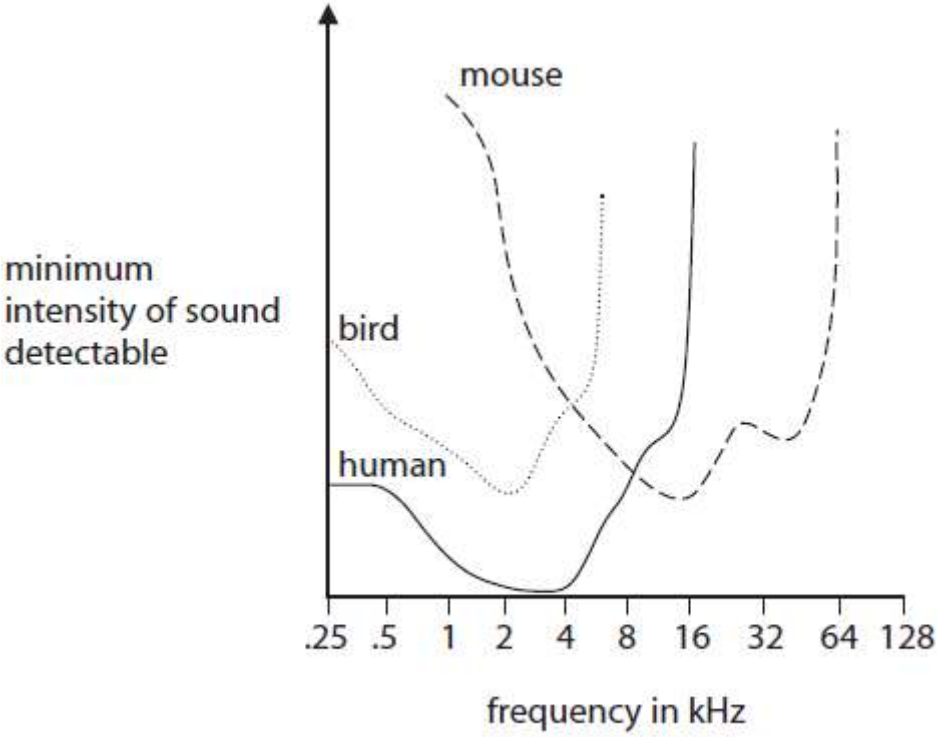

Figure 14

(i) Describe **two** differences between the hearing responses of the human and the mouse.

(2)

1 .....

.....

2 .....

.....

(ii) A farmer wants to use an alarm to scare away these birds.

State which frequency would be most effective.
Give the appropriate units.

(1)

frequency .....

(iii) State the reason for your choice of frequency in (ii).

(1)

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(Total for question = 4 marks)

Q4.

Diagram 1 shows a glass prism which can be used to turn an image the right way up.

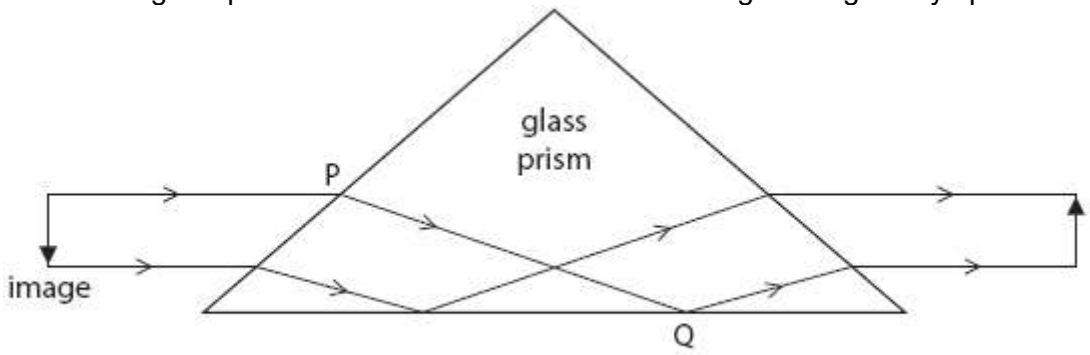

Diagram 1

(a) (i) In diagram 1, total internal reflection occurs at Q.
Explain why total internal reflection occurs at Q.

(2)

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(ii) The way in which the light changes direction at P is shown in diagram 2.
Mark on the diagram (*i*) for the angle of incidence and (*r*) for the angle of refraction for the ray of light shown.

(2)

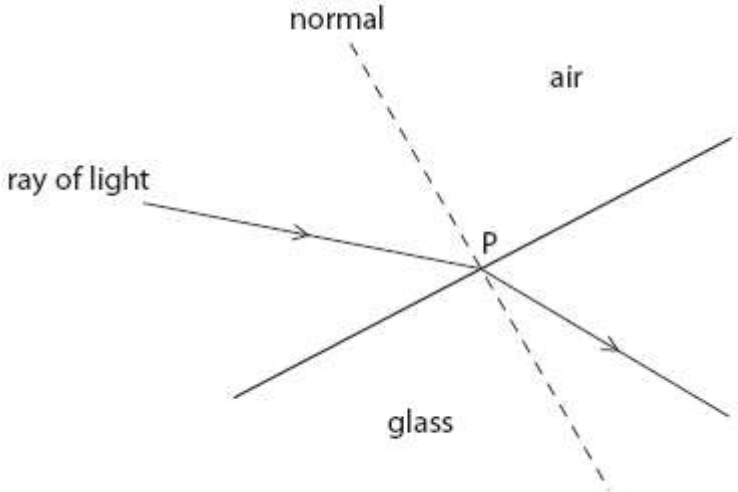

Diagram 2

(iii) Which of these is correct for the light as it enters the prism at P?
Put a cross (☒) in the box next to your answer.

(1)

☒ **A** frequency decreases

☒ **B** frequency increases

☒ **C** speed decreases

☒ **D** speed increases

(b) Light waves and sound waves are both used in the diagnosis and treatment of medical conditions

(i) A doctor uses an endoscope to look inside the body of a patient.

Explain how optical fibres are used in endoscopes.

You may draw a labelled diagram to help with your answer.

(3)

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(ii) Describe how ultrasound can be used as a medical treatment for illness or injury.

(2)

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(Total for Question = 10 marks)

Q5.

Figure 16 is an energy diagram for a sound wave incident on a sound-insulating board.

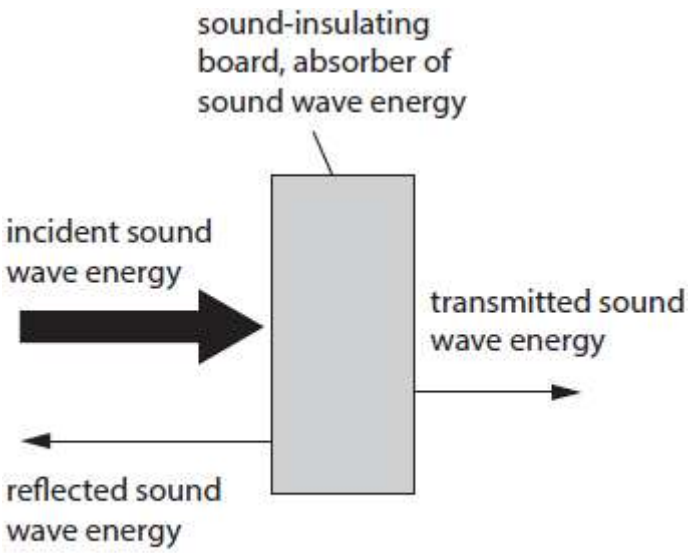

Figure 16

(i) The incident energy is 0.25 J.

The absorbed energy is 67% of the incident energy.
The reflected energy is 15% of the incident energy.
Calculate the amount of the transmitted energy.

(2)

transmitted energy = ... J

(ii) Give **one** way to reduce the percentage of energy transmitted through the sound-insulating board.

(1)

.....

.....

(Total for question = 3 marks)

Q6.

Figure 16 is an energy diagram for a sound wave incident on a sound-insulating board.

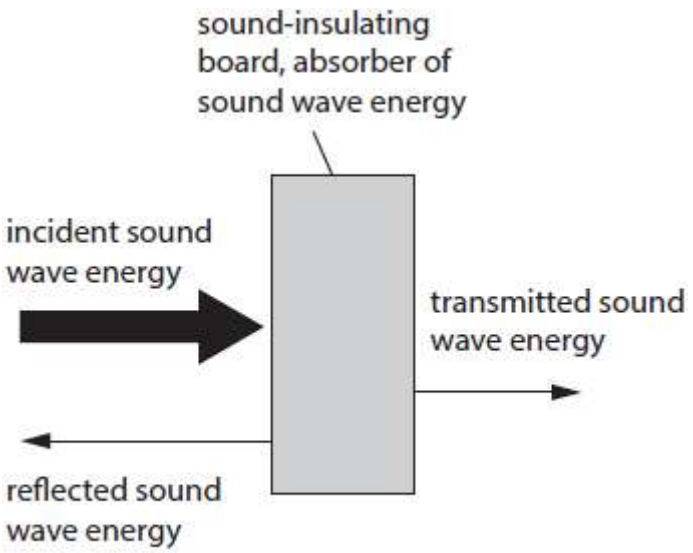

Figure 16

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The reflected energy is 15% of the incident energy.
Calculate the amount of the transmitted energy.

(2)

transmitted energy = ... J

(ii) Give **one** way to reduce the percentage of energy transmitted through the sound-insulating board.

(1)

.....

.....

(Total for question = 3 marks)

Q7.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Figure 9 shows two technicians, L and M, measuring the speed of sound in air.

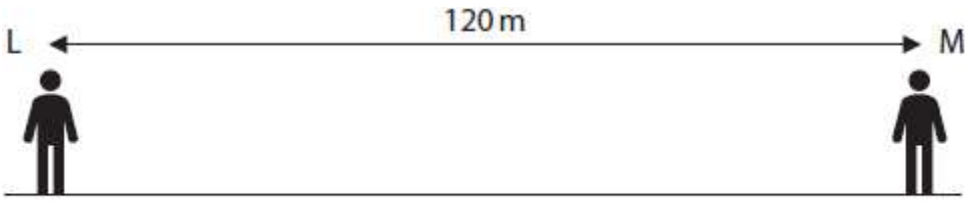


Figure 9

L fires a starting pistol.

M starts a stopwatch when first seeing the smoke from the starting pistol.

M stops the stopwatch when hearing the bang made by the starting pistol.

The distance between L and M is 120 m.

M's reaction time is 0.23 s.

The speed of sound in air is 330 m / s.

(i) Calculate M's reaction time as a percentage of the time sound takes to travel from L to M.

(3)

..... %

(ii) Which of these would improve the technicians' measurement of the speed of sound?

(1)

- ☐ **A** Use a firework 'banger' instead of the starting pistol.
- ☐ **B** Use a stop clock that measures time in minutes.
- ☐ **C** Increase the distance between **L** and **M**.
- ☐ **D** Decrease the distance between **L** and **M**.

(Total for question = 4 marks)

Q8.

A transducer can transmit and detect ultrasonic waves.

Figure 15 shows ultrasonic waves transmitted by the transducer on the bottom of a ship.

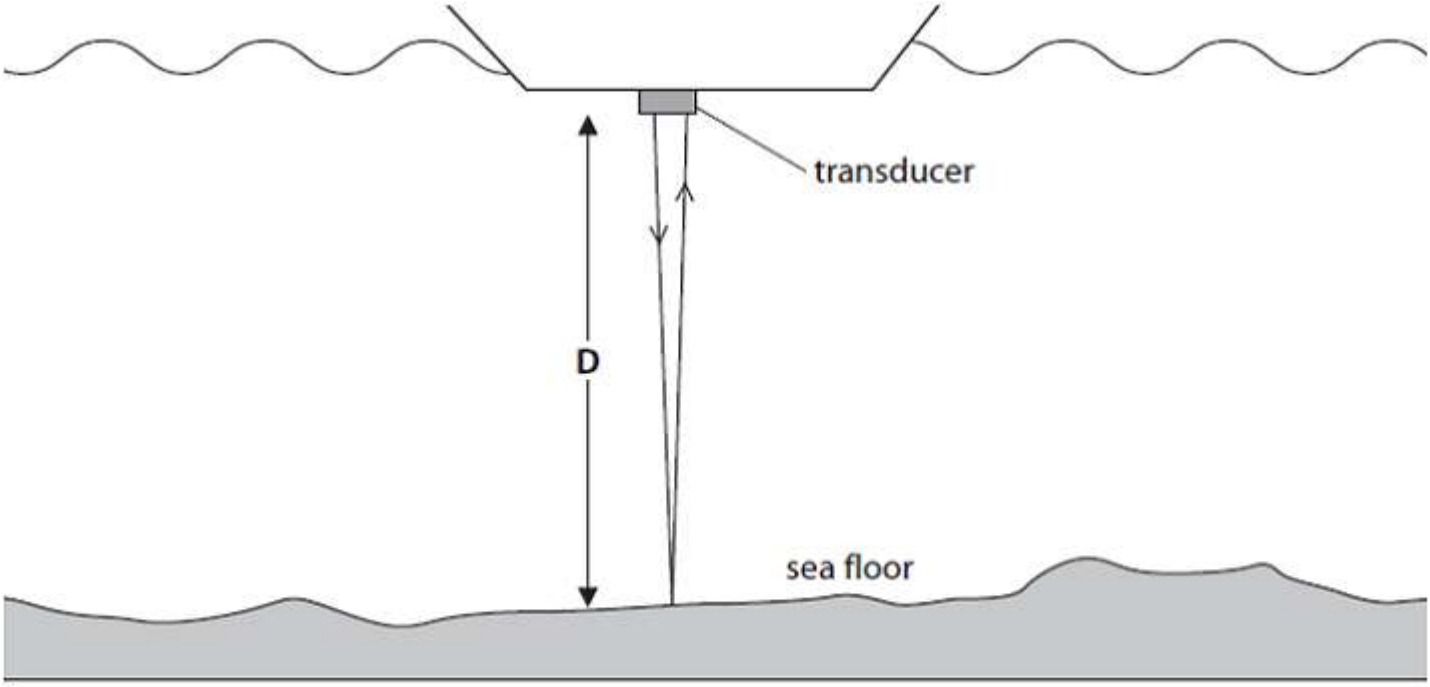


Figure 15

The waves reflect off the sea floor and are received back at the transducer.

The waves travel at 1500 m / s.

The time between transmission and reception is 48 milliseconds.

Calculate the depth of water, D, shown in Figure 15.

(2)

depth of water, D = ... m

(Total for question = 2 marks)

Q9.

The pulse returns to the bat after a time of 18 ms.

Calculate the distance from the bat to its prey.

(4)

distance = ... m

(Total for question = 4 marks)

Q10.

Figure 1 shows a bat and its prey.

Figure 1

The bat emits a high frequency sound pulse to locate its prey.

The speed of sound in air is 330 m / s.

The wavelength of the sound is 11 mm.

Calculate the frequency of the sound.

(2)

Use the equation

$$v = f \times \lambda$$

frequency = ... Hz

(Total for question = 2 marks)

Mark Scheme

Q1.

	Answer	Acceptable answers	Mark
	An explanation including: • Infrasound (1) Plus one from: • Decrease/change in amplitude is least (1) • can be detected/'heard' further away (1)	Marks are independent Stays the biggest/stays high. Has a bigger amplitude would travel the furthest/further	**(2)**

Q2.

Question number	Answer	Additional guidance	Mark
(i)	substitution (1) % difference = $\frac{(240 - 343)}{343} \times 100$ evaluation (1) (-) 30 (%)	 OR 343 – 240 in numerator award full marks for the correct answer without working allow 1 mark for division by 240 yielding 43% allow one mark for $\frac{240 \times 100}{343}$ = 70 %	**(2)**

Question number	Answer	Additional guidance	Mark
(ii)	explanation linking any two of: reaction time is significant (with 0.5s or less) (1) the reaction time will be different for each of the students (1) effects on reaction times (1) students are at different distances (from starting pistol) (1) anticipation of flash / bang (1)	 accept reaction time is large compared with travel time differences in perception / acuity of light and sound	(2)

Question number	Answer	Additional guidance	Mark
(iii)	explanation linking: use a (much) longer distance **OR** use electronic timer (1) with effect (1) reduces/eliminates the significance/impact of the reaction time **OR** gives a more manageable time to measure	 all stand the same distance from the starting pistol (1)	(2)

Q3.

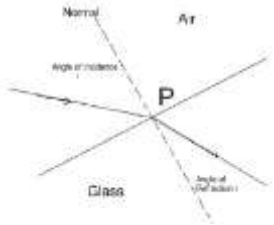
Question number	Answer	Additional guidance	Mark
(i)	Any two valid differences e.g. mice are (more) sensitive to higher frequencies (than humans) (1) humans are (more) sensitive to lower frequencies (than mice) humans have a lower intensity of sound (threshold) to just hear (1) mice have two minima on the graph, humans one (1)	 humans more sensitive (overall / for lower frequencies) two particularly sensitive frequencies	(2) grad

Question number	Answer	Additional guidance	Mark
(ii)	number **and unit** must both be correct to get the mark 2.5 kHz **OR** 2.5 kilohertz	 acceptable range 2.0 to 3.0 kHz 2.0 to 3.0 kilohertz	(1) cler

Question number	Answer	Additional guidance	Mark
(iii)	(this frequency corresponds with) most sensitive / the minimum (frequency) (1)		(1) grad

Q4.

	Answer	Acceptable answers	Mark
(a)(i)	An explanation linking: Angle (of incidence) in glass (1) greater than critical	Angle in air cannot be greater than 90° for 1 mark Glass has a higher refractive index than	**(2)**

	angle / 42° (1)	air for 1 mark	
(a)(ii)	 angle i from normal in air (1) angle r from normal in glass (1)	accept for 1 mark angle i in air <u>and</u> angle r in glass/ <u>both</u> angles measured from normal	**(2)**
(a)(iii)	☒ **C** speed decreases		**(1)**
(b)(i)	An explanation linking any three of the following: (Optical fibres) bend (1) some fibres carry light to the inside of the patient (1) some fibres transmit the reflected light (1) light passes up/down fibres by TIR (1) light is reflected inside the patient (1) image is analysed by computer (1)	Accept suitable labelling on a Diagram Image projected on a screen	**(3)**
(b)(ii)	Either Breaks/blasts/smashes (1) Kidney stones (1) or Energy absorbed (1) to help repair muscle tissue (1) or Use of gel (1) to prevent loss of intensity (1) or	bruising/clots increases blood flow Allow (1) mark for suitable diagnosis e.g. prenatal scan	**(2)**

Q5.

Question	Answer	Additional guidance	Mark
(i)	percentage transmitted (1) 100 – 67 – 15 18 (%) evaluation (1) $0.25 \times \frac{18}{100}$ 0.045 (J)	0.1675 or 0.0375 or 0.25 - 0.1675 - 0.0375 or 0.25 - 0.205 allow 82% calculated as 0.205 (J) for 1 mark 0.045 to any other power of ten scores 1 mark award full marks for the correct answer without working	**2** **AO2.1**

Question	Answer	Additional guidance	Mark
(ii)	one from increase the thickness (of it/ the absorber /the insulating board) (1) make the surface (of board) more reflective / make the board more absorbing (1) make board less/more dense (1) put an airgap in the insulating board (1) add another material with different density adjacent to board (1)	do not accept 'add more insulation' on its own ignore mass, surface area, volume, amount and size allow change the density (of the board)	**1** **AO3.1**

Q6.

Question	Answer	Additional guidance	Mark
(i)	percentage transmitted (1) 100 – 67 – 15 18 (%) evaluation (1) $0.25 \times \frac{18}{100}$ 0.045 (J)	0.1675 or 0.0375 or 0.25 - 0.1675 - 0.0375 or 0.25 - 0.205 allow 82% calculated as 0.205 (J) for 1 mark 0.045 to any other power of ten scores 1 mark award full marks for the correct answer without working	**2** **AO2.1**

Question	Answer	Additional guidance	Mark
(ii)	one from increase the thickness (of it/ the absorber /the insulating board) (1) make the surface (of board) more reflective / make the board more absorbing (1) make board less/more dense (1) put an airgap in the insulating board (1) add another material with different density adjacent to board (1)	do not accept 'add more insulation' on its own ignore mass, surface area, volume, amount and size allow change the density (of the board)	**1** **AO3.1**

Q7.

	Answer	Additional guidance	Mark
(i)	calculation of time of travel (1) (120/330 =) 0.36(36) (s) substitution (1) $\frac{0.23 \times 100}{0.36(36)}$ evaluation (1) 63 (%)	 ecf from MP1 for MP2&3 accept values that round to 64 or 63 accept values that round to 0.64 or 0.63 for 2 marks award full marks for the correct answer without working	(3) AO2

	Answer	Mark
(ii)	**C** Increase the distance between L and M. **A, B** and **D** are incorrect as these would not improve the technicians' measurement	(1) AO3

Q8.

Question number	Answer	Additional guidance	Mark
	recall and substitution (1) D = ½ v × t = ½1500 × 0.048 evaluation (1) 36 (m)	 award full marks for the correct answer without working award 1 mark only for 72 m (i.e. ½ ignored)	**(2) grad**

Q9.

	Answer	Additional guidance	Mark
	Two possible methods: selection (1) distance = speed x time substitution (1) (total distance =) 330 x 18 ($\times 10^{-3}$) evaluation of total distance (1) 5.9(4) (m) evaluation of distance from bat to prey (1) 3.0 (m)	 speed = distance / time 330 = d / 18($\times 10^{-3}$) 330x18($\times 10^{-3}$) scores mp1 and mp2 allow their distance divided by 2 for MP4 allow 2.97 (m) 5.94 scores 3 marks 5.94 to any other power of 10 scores 2 marks 2.97 or 3 to any other power of 10 scores 3 marks	**(4)** **AO2**

	OR selection (1) distance = speed x time division of time by 2 (1) 9 ($x10^{-3}$) substitution (1) (total distance =) 330 x 9 ($x10^{-3}$) evaluation of distance from bat to prey (1) 3.0 (m)	speed = distance / time 330 = d / 9($x10^{-3}$) 330x9($x10^{-3}$) scores mp1 and mp2 and mp3 allow 2.97 (m) 2.97 or 3 to any other power of 10 scores 3 marks award full marks for the correct answer without working	

Q10.

	Answer	**Additional guidance**	**Mark**
	substitution and rearrangement (1) (f =) $\frac{330}{11(x10^{-3})}$ evaluation (1) 30 000 (Hz)	 award full marks for the correct answer without working 30 or 300 or 3000 scores 1 mark	**(2)** **AO2**